MACHINE WHITE PAPERS

Hwacheon Hi-Tech 400 — Torque & Tumble

Mid-size CNC lathe with live tooling — turn, mill, drill, and thread in one setup. 12.5" OD × 49" L max turning.

Torque & Tumble · mid-size CNC lathe with live tooling

The Hi-Tech 400 is Hwacheon's heavy-duty box-way turning platform — the class of lathe you reach for when you're taking real chips off tough alloys. The 15" hydraulic chuck and 5.3" spindle bore let it accept the kind of stock most job-shop lathes won't clear, and the box-way construction keeps roughing cuts stable in Inconel and Duplex. Torque & Tumble handles wellhead spools, downhole tool bodies, valve stems, and any oilfield turning where rigidity and bore size matter more than cycle time.

What Torque & Tumble is for

Torque & Tumble is the machine that removes a setup. Live tooling on a heavy-duty box-way turning center means a part can be turned, milled, cross-drilled, and threaded in one chucking — 12.5" OD × 49" L, 12-station turret, Fanuc 18iTB. Every time a part comes out of the chuck and goes into another machine, you pay for it twice: once in handling and once in the concentricity you lose re-locating it. On parts with milled features referenced to a turned diameter, one setup is not a convenience, it's the tolerance strategy.

Pick this machine when

- The part is turned but has cross-holes, flats, slots, or milled features that must hold position to a turned datum.
- You've been running the part as two operations across two machines and the concentricity is marginal.
- Box-way rigidity matters — heavy interrupted cuts in duplex or Inconel at mid-size.
- Threaded tool bodies and subs where the thread and the milled feature share a reference.

> **When it's the wrong machine**
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> Pure turning at high RPM. The heavy-duty configuration tops out at 1,800 RPM, which is deliberate — this platform trades spindle speed for rigidity. Small high-volume turned parts cycle faster on Bearkat Blur. And if the part has no milled features at all, you're paying for live tooling you won't use.

From the floor

Live tooling is only worth having if the turret is treated as a real milling spindle, which means respecting its limits. Driven-tool cuts get lighter depths and more passes than the same feature would get on a VMC — the turret is rigid for a lathe, not rigid like a 25 HP BT50 column. Buyers who've been burned by live tooling usually got burned by someone pretending that distinction doesn't exist.

Pick your industry

Hwacheon Hi-Tech 400 for Oil & Gas

Full spec sheet, typical oil & gas parts, alloys, tolerance expectations, documentation approach.

Hwacheon Hi-Tech 400 for Aerospace

Full spec sheet, typical aerospace parts, alloys, tolerance expectations, documentation approach.

Hwacheon Hi-Tech 400 for Military & Defense

Full spec sheet, typical defense parts, alloys,

Hwacheon Hi-Tech 400 for Industrial & General Manufacturing

Full spec sheet, typical industrial parts, alloys, tolerance expectations, documentation approach.

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Have a project?